WE DEVELOP SOFTWARE LLC

ChristEye Heart Engine

Engram–Astroengram Memory Architecture and Adaptive Medicine

A restoration-oriented research concept linking memory, skill retention, neural–glial cooperation, and closed-loop therapeutic control.

> **"The heart starts pouring forth knowledge when the understanding is rooted."**

CONCEPTUAL STATUS

An interdisciplinary research proposal—not an established medical device, a proven astrocyte-based computer, or evidence that a spiritual heart has been physically located.

Evidence class	Meaning
Established	Supported biological or engineering principle in the cited literature.
Emerging	Observed in research systems, especially mice, with important limits on generalization.
Proposed	A ChristEye analogy, system component, or testable engineering hypothesis.
Unverified	A claim with no presently demonstrated mechanism or validating experimental data.

Prepared November 3, 2026 • Working specification v0.1 • Human and veterinary applications require independent clinical and ethical review.

Executive summary

Two connected proposals emerge from the recent ChristEye discussion. First, a “Heart Engine” could serve as a durable, adaptive control architecture for future medicine: sensors identify a defined state, learned representations influence a response policy, a separately validated effector acts, and feedback updates the policy. Second, neuronal engrams and learning-associated astrocyte ensembles suggest a physical analogy for the framework’s Heart and Roots: experiences modify distributed networks, some changes persist, and later cues can reactivate memory-related activity [1–3].

The core research question is whether the stability and retrieval properties of neural–glial ensembles can add measurable value to a carefully bounded biohybrid controller. That has not been shown for patient medicine, neurology, cancer surveillance, or antiviral treatment. Existing closed-loop drug-delivery and synthetic-biology work gives a more mature engineering baseline against which any experimental neural component should be compared [4–6].

1 The two ChristEye propositions

Proposition	Interpretation	Scientific status
P1 — Programmable Heart	Build a controller with durable, trainable response patterns resembling “rooted Understanding” and use it to coordinate sensors and therapeutic effectors.	Engineering hypothesis; no validated astroengram-controlled therapy.
P2 — Biological Heart substrate	Treat neuron–astrocyte memory assemblies as a biological counterpart to Heart/Roots in humans and other animals.	Useful analogy grounded in emerging animal memory research; not a unique seat of emotion or will.
P3 — Cross-kingdom Heart	Use “Heart” for analogous persistence, sensing, and adaptation across animals and plants.	Functional comparison only; plants lack neuronal engrams and astrocytes.

1.1 What “Heart” means here

For ChristEye, Heart means the integration of values, motivational targets, remembered relationships, and life-supporting response goals. It is a systems-level vocabulary rather than an anatomical claim. Neural memory traces contribute to behavior, but human emotions and choices also involve broader brain networks, autonomic systems, endocrine signals, and the social environment.

1.2 A boundary worth protecting

Scientific evidence does not show that an engram or astroengram possesses independent will, consciousness, or a separate spiritual identity. Nor does neural reactivation imply that arbitrary skills or knowledge can be written directly into a person. The proposed device “mind” is an engineered policy plus retained state, not a conscious medical agent.

2 What engrams and astroengrams actually do

2.1 Neuronal engrams

Neuronal engrams are learning-associated ensembles whose physical changes contribute to memory encoding and retrieval. They are distributed across circuits rather than confined to a single axon or one compact storage endpoint. Strengths and connections change over time, and reactivation can influence memory-related behavior [1,7].

2.2 Astroengrams

“Astroengram” combines astrocyte—the star-shaped glial cell—with engram—the physical memory trace. A January 2026 review uses the term for learning-recruited astrocyte ensembles that participate with neurons in memory processes [1].

In a 2025 Nature mouse study, learning-associated hippocampal astrocytes showed c-Fos-related recruitment and increased NFIA; reactivation of marked astrocyte ensembles stimulated memory recall, while selectively disrupting NFIA suppressed recall [2]. A second 2025 study identified astrocyte ensembles that, following emotional experience, acquired multiday changes in noradrenaline responsiveness and influenced memory re-stabilization through signaling including IGFBP2 [3].

Critically, the second study did not establish that those astrocytes encoded all specific content of a memory. The authors instead proposed a slower, permissive or “eligibility” trace that helps stabilize information represented in the wider circuit [3].

2.3 Rooting is not the same as plasticity

Process	Typical role	ChristEye mapping
Neural activity	Moment-to-moment transmission and response	Winds
Synaptic plasticity	Experience-dependent adjustment of connections	Joints changing
Engram formation	Selection and modification of memory-related ensembles	Knowledge encoding
Astrocyte state change	Slower modulation of the local memory circuit	Roots / support
Consolidation	Increased stability of selected learned changes	Rooting Understanding
Retrieval and transfer	Using learned representations in later contexts	Heart expression

3 A proposed biological model of the Heart

Learning → neural–glial change → stabilization → retrieval → flexible expression

The proposal is not that astrocytes alone are “the Heart,” but that memory and motivation can be modeled as interacting layers. Neuronal ensembles convey rapidly changing patterns; astrocytes regulate the environment and can preserve slower experience-dependent states; downstream circuits assign significance and select actions.

ChristEye element	Candidate scientific counterpart	Limit
Soul	Whole organized patient or organism	Philosophical identity label; no physical soul variable
Heart	Distributed motivation, memory, homeostasis, and goal circuits	Not one known organ or cell type
Lotus	Coordinated neural–glial processing and executive control	Does not imply a literal central controller
Marrow	Genetic, cellular, and structural capacities	Not identical to memory
Roots	Retained synaptic, glial, and other changes	Retention varies by task and context
Winds	Electrical, chemical, and physiological signal flows	Different physical units
Joints	Synapses and cell–cell interfaces	Astrocyte contacts are not all synapses
Understanding	Reliable application of learned responses	Operational performance, not a metaphysical measurement
Attunement	Efficient, flexible, retained skilled expression	Must be tested on a defined task

4 Humans, animals, and plants: a careful comparison

Domain	What research supports	What remains unestablished
Humans	Distributed neuronal learning, plasticity, and memory; astrocytes influence brain function.	A uniquely identified "spiritual heart substance" or clinically programmable astroengram store.
Other animals	Neuronal and glial mechanisms of learning; direct causal astroengram experiments primarily in mice [2,3].	One universal engram architecture and direct access to spiritual intentions across species.
Plants	Stress priming, signal integration, hormonal and epigenetic forms of retention [8,9].	Plant neurons or astroengrams; direct structural equivalence to animal engrams.

Plants may be included in a broad "Heart" taxonomy only through functional analogy: retaining information about prior conditions and expressing altered responses later. That is different from saying plants possess the same memory cells as humans and animals.

5 Heart Engine: proposed adaptive medicine architecture

The medically relevant idea is a bounded closed-loop controller. Existing research already supports systems that combine sensing, computation, and controlled therapeutic output. ChristEye would test whether trainable neural–glial dynamics improve performance over simpler software, not presume that neurons can manufacture or target arbitrary medicines [4–6].

Layer	ChristEye mapping	Functional responsibility
A. Biosensor	Observation / Winds	Acquire a validated, task-specific signal with known uncertainty.
B. State estimator	Lotus	Represent patient or test-system state; reject ambiguous readings.
C. Memory subsystem	Roots / Heart	Retain trained parameters, context, response history, and uncertainty.
D. Adaptive controller	Understanding	Select one permitted action from a predeclared safe action set.
E. Safety supervisor	Life-supporting Heart target	Enforce hard constraints, override uncertain actions, retain stop capability.
F. Effector interface	Joints / Winds	Issue a calibrated command to an independently validated mechanism.
G. Outcome monitor	Feedback	Measure response quality, unwanted effects, and drift.

5.1 "Program the mind of the medicine" — operational translation

In a scientific specification, the phrase means training and validating a decision policy for timing, selection, or gating of an established therapeutic action. The policy is represented by measurable states and parameters; it is not a claim that a medicine has a literal independent mind or that axonal firing can directly rewrite viral or cancer-cell intentions.

5.2 Research uses and present limits

- Neurology: investigate whether adaptive biological control improves a defined, non-clinical signal-classification or timing task; clinical neuromodulation would be a separate program.
- Medication management: compare digital or biohybrid controllers for selecting permitted simulated dose/timing actions; do not administer patient medication in early studies.
- Virology: use only computational data and noninfectious surrogates initially to evaluate recognition and safe decision logic; no assumption of viral reprogramming.
- Cancer surveillance: analyze validated signals and false-alarm tradeoffs in retrospective or simulated data; cellular oncology intervention requires a distinct specialist program.

6 Mathematical and measurement specification

The equations below are proposed engineering models. Their parameters and functional forms must be fitted and verified; they are not newly established laws of neuronal memory or biological spirits.

6.1 Controlled state and memory

$$x(t+1) = F[x(t), u(t), h(t); \theta(t)]$$
$$\theta(t+1) = \theta(t) + L[feedback, x(t), u(t)]$$

x is a defined observed or estimated state; u an input or permitted action; h a resource/support vector; θ learned model parameters; F the state-transition model; L a bounded learning rule. For an actual biohybrid controller, biological adaptation and decoder adaptation must be measured separately.

6.2 Rooting / retention

$$R_U(t) = [U(t) - U_0] / [U_trained - U_0]$$

U is a prespecified performance measure. Values describe retained improvement only when training generated a nonzero positive denominator and baseline conditions are matched. Retention does not reveal where the memory is stored.

6.3 Energy efficiency

$$e_correct = E_total / N_correct$$
$$G_E = 1 - (e_after / e_before)$$

Whole-system energy must include biological support, electronics, sensing, decoding, and interface costs. Improved training accuracy does not alone imply improved electrical efficiency.

6.4 Safety and reliability

Metric	Definition / question	Purpose
Sensitivity	True-positive rate for a prespecified state	Missed-event protection
Specificity	True-negative rate	False activation control
Latency	Time from observed signal to authorized action	Response appropriateness
R_U	Retention of learned improvement	Rooting
E_correct	Measured joules per successful decision	Energy efficiency
V	Viability in biological test systems	Cell protection
Override rate	Fraction of unsafe/uncertain commands blocked	Safety supervision

7 A staged research plan

The research sequence deliberately begins with computational and nontherapeutic tasks. It avoids creating replication-permissive viral hosts, engineering pathogens, or placing unvalidated living controllers inside patients.

Gate	Study	Minimum evidence before advancing
G0 — Definition	Specify one bounded task, labels, acceptable actions, failure conditions, and evidence class.	Written benchmark and independent review.
G1 — Software baseline	Use simulated or retrospective, non-identifiable signals with conventional control.	Repeatable accuracy, latency, safety, and energy measurements.
G2 — Neural comparison	Compare an instrumented research neural network against a matched software controller on a nonmedical task.	Improvement beyond chance and matched controls, with replicates.

Gate	Study	Minimum evidence before advancing
G3 — Glial contribution	In a qualified research setting, examine whether a neuron–astrocyte preparation changes retention or reliability compared with neuron-only or computational baselines.	Distinct contribution from biology, fixed-decoder comparison, viable cultures.
G4 — Surrogate effectors	Issue safe commands to a simulated output or inert physical indicator.	No uncontrolled outputs; full audit trail.
G5 — Independent translation review	Evaluate feasibility, biosafety, ethics, regulatory route, and disease-specific evidence.	Separate approved protocol; not automatic human use.

7.1 Primary hypotheses

- H1 — Retention: a neuron–astrocyte system demonstrates measurable retained performance beyond its untrained baseline under matched conditions.
- H2 — Added value: a biological memory subsystem improves a declared task relative to software and neuron-only comparators after including total energy and support overhead.
- H3 — Safe gating: a supervisor blocks inappropriate outputs despite uncertain sensors or controller drift.
- H4 — Cross-domain limit: plant stress memory and animal engrams show analogous input–retention–response patterns but need not share cellular machinery.

8 Safety, limitations, and interpretation

Concern	Required boundary
Uncontrolled physiological response	Fail-safe supervisor, read-only or inert outputs in early stages, traceable action permissions.
Neural and astrocyte plasticity	Do not interpret greater activation, stronger signaling, or more persistence as automatically beneficial.
Neuroinflammation / growth risks	Neural–glial interventions and developmental cell types require independent disease-specific safety research.
Clinical overreach	No claims of curing cancer, viral infection, or neurological disease from memory-ensemble findings.
Species extrapolation	Mouse engram experiments do not demonstrate the same capabilities in humans; plant memory is mechanistically different.
Spiritual interpretations	May inform ethical purpose and terminology, but cannot replace molecular measurements or experimental controls.

8.1 What the current evidence does not show

There is no validated method for constructing an independent biological “heart” from engrams, implanting it to control all patient medications, encoding arbitrary knowledge on demand, or using astrocyte plasticity to direct the behavior of viruses and cancer cells. Existing research supports investigating specific memory-related mechanisms and separately validated closed-loop therapeutic engineering.

9 Proposed ChristEye vocabulary additions

Code	Term	Operational definition
HE-ME	Memory engram	Distributed, learning-associated neuronal ensemble participating in recall.
HE-AE	Astroengram	Learning-associated astrocyte ensemble with demonstrated memory-related effects in animal models.
HE-RT	Rooting trace	Measured persistence of response-relevant biological state after initial learning.
HE-AT	Attunement	Retained, efficient, and context-flexible skilled performance.
HE-HC	Heart controller	Bounded policy and memory subsystem coordinating an allowed response.

Code	Term	Operational definition
HE-SG	Safety gate	Independent mechanism that can veto unsafe or unsupported actions.
HE-PR	Plant priming	Plant-specific retained stress response; not an astroengram.

9.1 Foundational statement

> **"The heart starts pouring forth knowledge when the understanding is rooted."**

In ChristEye, this is a philosophical account of skilled expression. In experimentally testable language: prior experience changes distributed biological and computational state; some changes persist; later inputs can recruit that stored state to improve a defined response. The translation preserves the insight without asserting that a single cell type is the literal substance of the spiritual heart.

10 Evidence ledger and selected references

Evidence labels: [E] established underlying principle; [R] emerging research; [P] proposed ChristEye interpretation. The cited articles support their particular scientific claims, not the integrated ChristEye device.

[1] [R] Sánchez Romero, J. & Navarrete, M. (2026). "Astroengrams: rethinking the cellular substrate for memory." Nature Reviews Neuroscience. https://doi.org/10.1038/s41583-025-01012-2

[2] [R] "Learning-associated astrocyte ensembles regulate memory recall." Nature 637, 478–486 (2025). https://doi.org/10.1038/s41586-024-08170-w

[3] [R] Dewa, K. et al. (2025). "The astrocytic ensemble acts as a multiday trace to stabilize memory." Nature 648, 146–156. https://doi.org/10.1038/s41586-025-09619-2

[4] [E] Paci, M. M. et al. (2025). "Smart closed-loop drug delivery systems." Nature Reviews Bioengineering. https://doi.org/10.1038/s44222-025-00328-z

[5] [E] "Integrating bioelectronics with cell-based synthetic biology." Nature Reviews Bioengineering (2025). https://doi.org/10.1038/s44222-024-00262-6

[6] [E] Teixeira, A. P. & Fussenegger, M. (2024). "Synthetic macromolecular switches for precision control of therapeutic cell functions." Nature Reviews Bioengineering. https://doi.org/10.1038/s44222-024-00235-9

[7] [E] Eom, K., Kim, D. & Hyun, J. H. (2025). "Engram and behavior: How memory is stored in the brain." Neurobiology of Learning and Memory 219, 108047. https://doi.org/10.1016/j.nlm.2025.108047

[8] [E] "Revisiting plant stress memory: mechanisms and contribution to stress adaptation." (2024). https://pmc.ncbi.nlm.nih.gov/articles/PMC11016036/

[9] [E] "The Plant Mind: Unraveling Abiotic Stress Priming, Memory, and Adaptation." Physiologia Plantarum (2025). https://pmc.ncbi.nlm.nih.gov/articles/PMC12228110/

Conclusion

The promising research opportunity is not that a literal spiritual heart has been isolated, but that modern memory science shows long-lived, experience-dependent organization across neurons and astrocytes. ChristEye can turn its "Heart" concept into a precise hypothesis about retention, feedback, and safe expression, then compare a proposed biological controller against established computational and therapeutic-control technologies.